

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Short Answer Text (High)
Assessment Tool Weightage: 40%

QUESTIONS		
Q. No	Question	Bloom's Level
1	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember
2	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand
3	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand
4	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand
5	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q1. Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.

Schema

A **schema** is the overall logical structure or blueprint of a database. It defines how data is organized, including tables, attributes, relationships, constraints, and data types. A schema is designed once and changes very rarely.

Example:

Student (Student ID , Name, Department, CGPA)

This defines the structure of the Student table.

Instance

A **database instance** is the actual data stored in the database at a particular point in time. Unlike the schema, the instance changes whenever records are inserted, updated, or deleted.

Example:

Student_ID	Name	Department	CGPA
101	Rahul	CSE	8.7
102	Priya	ECE	9.1

The above records represent one instance of the Student database.

Difference between Schema and Instance

Schema

Blueprint of the database

Changes rarely

Defines structure

Created during database design

Instance

Actual data stored in the database

Changes frequently

Represents current contents

Updated during database operations

Types of Schema

1. Logical Schema

- Describes the logical structure of the database.
- Defines tables, attributes, relationships, and constraints.
- Independent of physical storage.

Example: Student(StudentID , Name, Department, CGPA)

2. Physical Schema

- Describes how data is physically stored on storage devices.
- Includes indexing, file organization, and storage allocation.
- Managed by the DBMS for efficient retrieval.

3. View Schema (External Schema)

- Defines how different users view the database.
- Provides customized views and enhances security by restricting access.

Example: A faculty member may view only **Student_ID**, **Name**, and **CGPA**, while the administrator can access all student details.

Q2. Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.

Entities and Attributes

1. Customer

- Customer_ID (Primary Key)
- Name
- Address
- Phone_Number
- Email

2. Account

- Account_Number (Primary Key)
- Account_Type
- Balance
- Opening_Date

3. Branch

- Branch_ID (Primary Key)
- Branch_Name
- Location

4. Loan

- Loan_ID (Primary Key)
- Loan_Type
- Loan_Amount
- Interest_Rate

5. Employee

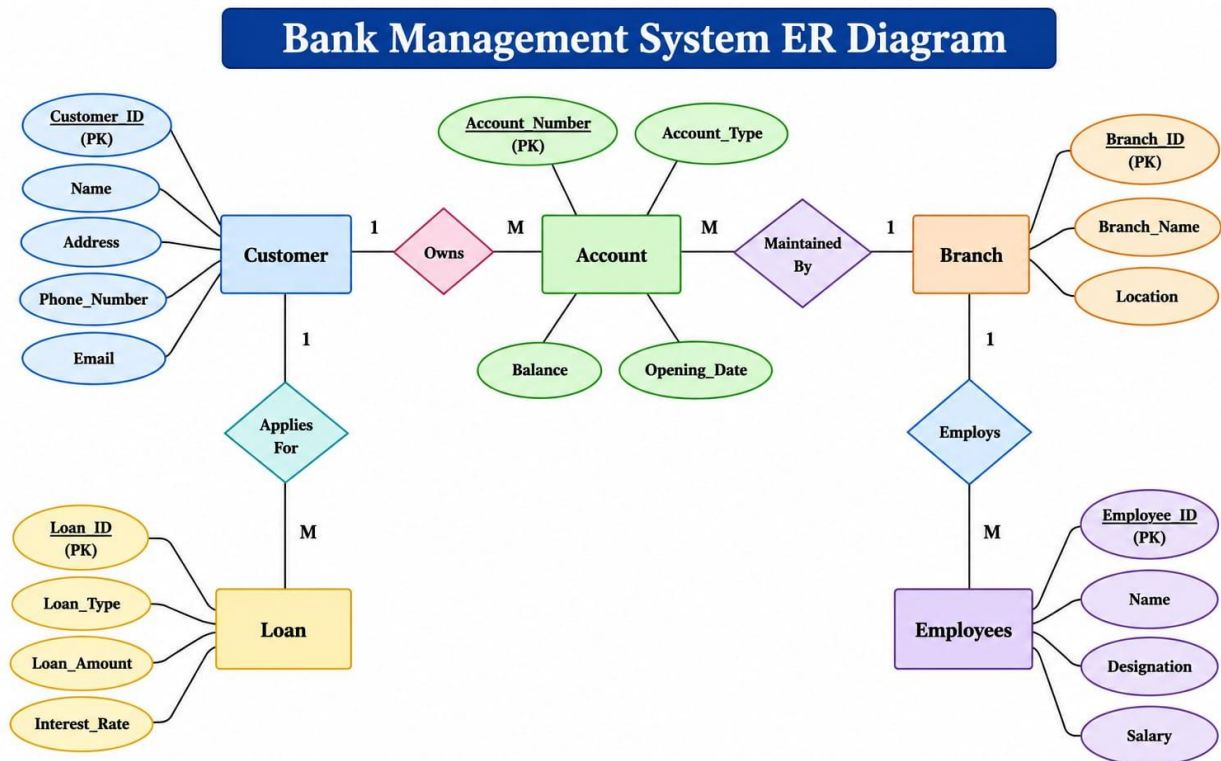
- Employee_ID (Primary Key)
- Name
- Designation
- Salary

Relationships

- A **Customer** owns one or more **Accounts**.
- An **Account** is maintained by one **Branch**.
- A **Customer** may apply for one or more **Loans**.
- A **Branch** employs many **Employees**.

- An **Employee** works in one **Branch**.

ER Diagram



Q3. Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.

Extended ER (EER) Model

The **Extended Entity-Relationship (EER) model** is an advanced version of the ER model that supports more complex database designs. It introduces concepts such as **inheritance**, **specialization**, **generalization**, **aggregation**, and **categories**, making it suitable for real-world applications.

Features of EER Model

- Supports inheritance.
- Represents superclass and subclass relationships.
- Models complex real-world scenarios.
- Reduces redundancy through hierarchical design.
- Improves database flexibility and scalability.

Generalization

Generalization is a **bottom-up approach** where multiple similar entities are combined into a single higher-level entity called a **superclass**.

Example:

Savings Account

Current Account

↓

Account

Both Savings Account and Current Account share common attributes such as Account Number and Balance, so they are generalized into the superclass **Account**.

Specialization

Specialization is a **top-down approach** where one superclass is divided into multiple subclasses based on distinguishing characteristics.

Example:

Employee

|

|

|

Doctor

Nurse

Doctor and Nurse inherit common Employee attributes while having their own specific attributes.

Aggregation

Aggregation represents a **relationship between a relationship and another entity**. It treats a relationship as a higher-level entity.

Example:

Employee ---- Works_On ---- Project

|

Managed By

|

Department

Aggregation helps represent complex relationships involving multiple entities.

Q4. Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.

Entities**Patient**

- Patient_ID (Primary Key)
- Name
- Age

- Gender
- Address

Doctor

- Doctor_ID (Primary Key)
- Name
- Specialization

Department

- Department_ID (Primary Key)
- Department_Name

Appointment

- Appointment_ID (Primary Key)
- Date
- Time

Medical Record (Weak Entity)

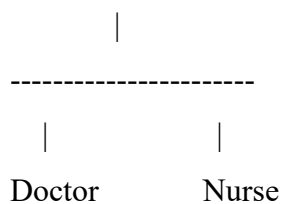
- Record_Number (Partial Key)
- Diagnosis
- Prescription

Medical Record depends on the Patient entity and cannot exist independently.

Inheritance (Specialization)

Employee

(Employee_ID, Name)

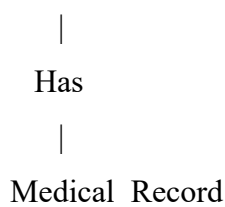


Doctor and Nurse inherit Employee attributes while adding their own specialized information.

Weak Entity

Patient

(Patient_ID)

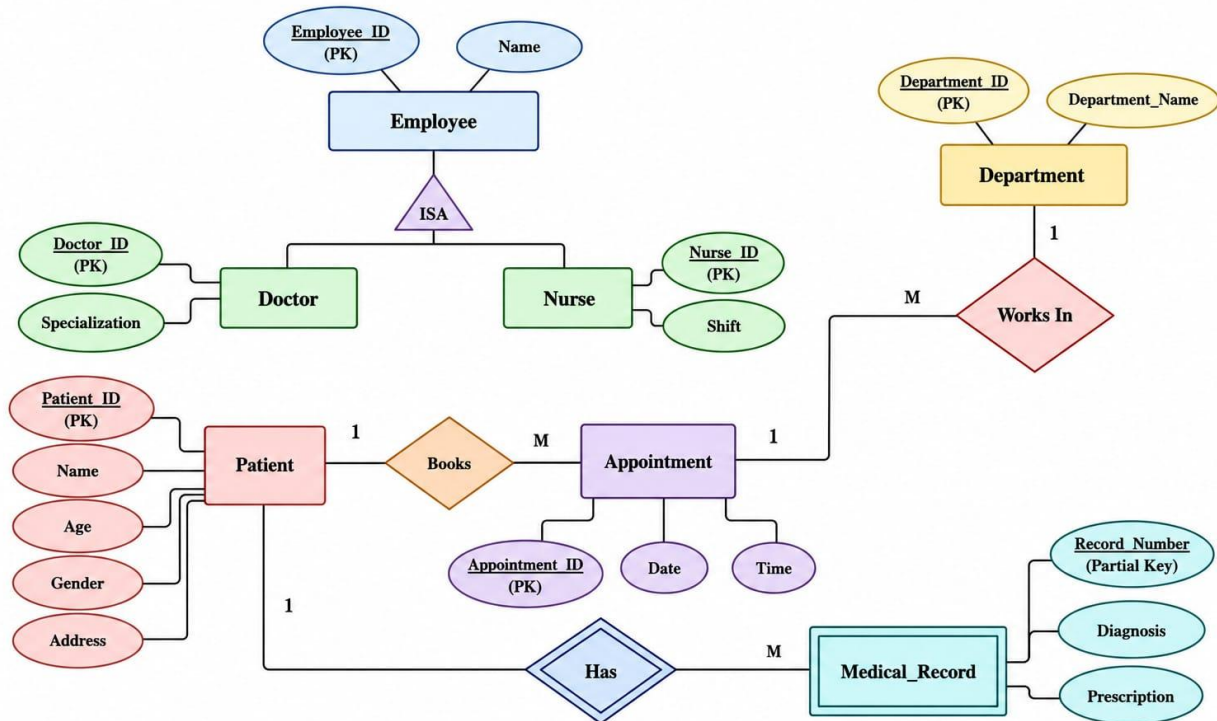


(Record_No, Diagnosis, Prescription)

The Medical Record is identified using **Patient_ID + Record_Number**.

EER Diagram

Hospital Management System EER Diagram



Q5. Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.

Database Administrator (DBA)

A **Database Administrator (DBA)** is responsible for managing, maintaining, securing, and optimizing the organization's databases. The DBA ensures that databases remain reliable, secure, and available while supporting efficient data storage and retrieval.

Responsibilities of a DBA

1. Database Installation and Configuration

- Installs, configures, and upgrades database management systems.

2. Database Design

- Creates database structures, tables, indexes, and relationships.

3. Security Management

- Controls user authentication, authorization, and access privileges to protect sensitive data.

4. Backup and Recovery

- Performs regular database backups and restores data during system failures or

disasters.

5. Performance Monitoring and Optimization

- Monitors database performance, optimizes queries, and manages indexes for efficient operation.

6. Data Integrity

- Ensures data consistency, accuracy, and enforcement of integrity constraints.

7. Storage Management

- Manages database storage, capacity planning, and efficient resource utilization.

8. Troubleshooting and Maintenance

- Identifies and resolves database issues while performing routine maintenance and updates.

9. Disaster Recovery Planning

- Develops recovery strategies to ensure business continuity after unexpected failures.

10. Support for Users and Developers

- Assists users and developers in database-related tasks and ensures smooth database operations.